

Appendix for 'SCI-Mamba: Unsupervised Learning based Low-Light Image Enhancement for Non-Cooperative Spacecraft'

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I. INFERENCE EXPERIMENT

This appendix provides the qualitative comparison results omitted from the main manuscript due to space constraints. The main manuscript focuses on the quantitative comparison of inference efficiency and resource consumption; the visual comparisons of enhancement outputs are presented here. Subsection I-A tests a synthetic virtual orbital picture from *Space Dark-1.0*, Subsection I-B a darkroom hardware-in-the-loop capture of a scaled satellite model, and Subsection I-C an authentic on-orbit spacecraft picture. Subsection I-D further tests the SPARK 2021 dataset [1], which contains space imagery not used in training. In these comparisons, SCI-Mamba generally brightens the spacecraft region while keeping the background and structural details relatively stable, whereas the baseline methods show different behaviors across scenes, including conservative enhancement, over-exposure, background noise amplification, and color or block artifacts. Detailed observations are given in the following subsections.

A. Test on Virtual Pictures

We first conduct evaluations on the synthetic image subset of *Space Dark-1.0*. Figure 1 visualizes the restoration outputs of all competing algorithms under simulated orbital illumination conditions.

For moderately dim synthetic scenes with hidden satellite shown in Figure 1, the raw input already outlines spacecraft hull and solar panels, yet shadowed zones, faint textures, and tiny bottom components remain obscured. Transformer-based methods Uformer and LLFormer produce overly conservative outputs that leave dark-region details unresolved. CNN-based unsupervised methods (SCI++ [2], Zero-DCE++ [3], RUAS [4]) improve the visibility of the spacecraft to varying degrees, but their enhancement of shadowed structures remains limited or spatially imbalanced. LLFlow [5] and UHDformer [9] suffer from over-enhancement artifacts, while ECMamba [7] produces an overly intense response with noticeable highlight saturation, while WalMaFa [8] also exhibits less balanced illumination restoration. By contrast, SCI-Mamba selectively brightens spacecraft regions without lifting the overall brightness of dark cosmic backgrounds. Solar-panel outlines and hull boundaries become more distinguishable, while small-scale

structural features are better preserved. This observation verifies that our model avoids naive global stretching; instead, it adaptively modulates illumination according to spatial texture and brightness statistics, retaining natural target-background radiometric contrast while boosting target visibility.

B. Test on Real Pictures

We proceed to hardware-in-the-loop darkroom tests using scaled satellite models. Real captured data contains more complex composite degradation including sensor noise, uneven lighting, local specular highlights and faint textures, which provides stricter testing of noise suppression, color correction and structural preservation capacity. Comparative visual result is illustrated in Figure 2.

Figure 2 presents lower light hardware-captured samples with extremely low hull luminance. Under such harsh conditions, most baseline methods tend to exaggerate background noise during brightness compensation. SCI++, Uformer, and LLFormer provide relatively conservative enhancement, leaving some fine spacecraft structures insufficiently visible. Zero-DCE++, ECMamba, WalMaFa, LLFlow, and UHDformer [9] enhance the target intensity, but simultaneously amplify background noise to varying degrees; LLFlow and UHDformer additionally suffer severe purple color shifts and local over-exposure that break visual plausibility. SCI-Mamba steadily improves satellite brightness while preserving continuous edge and texture information, without widespread noise proliferation across the background. The method demonstrates strong adaptability to real sensor-induced noise and uneven illumination artifacts beyond synthetic datasets.

C. Test with On-Orbit Spacecraft Pictures

Real orbital footage contains more complex degradation sources including local specular reflections, shadow occlusion, uneven background luminance and sensor noise, which directly reflect the practical deployment value of enhancement pipelines for spaceborne visual perception. Comparative result on authentic on-orbit imagery is shown in Figure 3.

For moderately illuminated orbital targets (Figure 3), the raw image already contains a recognizable spacecraft body, but shadowed regions exhibit low contrast and several local structural details remain difficult to distinguish. SCI++ and Uformer provide relatively conservative enhancement, leaving some shadowed regions insufficiently illuminated. RUAS,

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Fig. 1. Visual Comparison on Synthetic Virtual Orbital Picture - with Hidden Satellite



Fig. 2. Visual Comparison on Darkroom Captured Satellite Picture

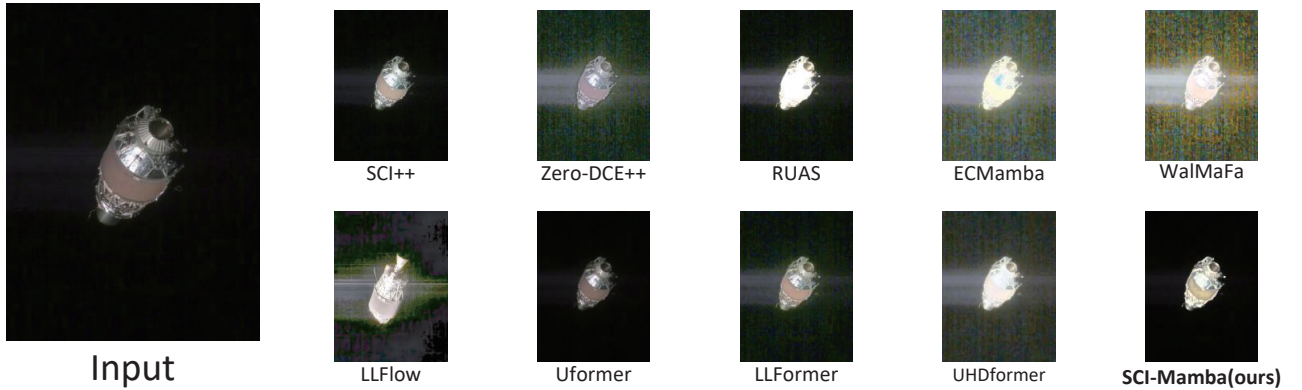


Fig. 3. Visual Comparison on Medium-bright Real On-orbit Spacecraft Picture

ECMamba and UHDformer overexpose reflective satellite surfaces and cause highlight clipping. Zero-DCE++, WalMaFa, and LLFormer amplify background noise while enhancing the spacecraft. LLFlow generates obvious strip and block artifacts that disrupt continuous structural representation. In contrast, SCI-Mamba brightens satellite regions while fully retaining hull contours and surface textures, avoiding highlight saturation on reflective components and effectively restraining background noise growth. The stable illumination correction and structural retention capacity of our method supply high-quality preprocessed inputs for downstream tasks such as tar-

get detection, pose measurement and component identification.

D. Generalization Test on the SPARK 2021 Dataset

To evaluate the generalization capability of SCI-Mamba on unseen space imagery, we further conduct experiments on the SPARK 2021 dataset [1]. The images in the SPARK 2021 dataset are generated in a realistic space simulation environment with highly diverse sensing conditions, including extreme and challenging scenarios across different orbits, background noise, low signal-to-noise ratios (SNRs), and the

high image contrast characteristic of real space imagery. The corresponding visual comparison is presented in Figure 4.

As shown in Figure 4, the spacecraft is located against a predominantly dark space background, while a bright Earth region appears in the lower-right portion of the image. Due to insufficient illumination, the spacecraft body and solar panels exhibit relatively low visibility. Different methods exhibit noticeably different enhancement characteristics. SCI++ and Uformer provide relatively conservative enhancement, leaving several structural details of the spacecraft insufficiently visible. Zero-DCE++, ECMamba, WalMaFa, LLFlow, LLFormer, and UHDformer increase the overall brightness to varying degrees, but also amplify the dark background. In particular, LLFlow and LLFormer introduce noticeable background noise and color distortion. RUAS significantly brightens the spacecraft, but simultaneously over-enhances the bright Earth region, resulting in partial highlight saturation.

In contrast, SCI-Mamba selectively enhances the spacecraft while maintaining the low-intensity characteristics of the surrounding space background. The spacecraft body and solar panels become more distinguishable without introducing extensive background amplification, while the high-intensity Earth region remains relatively stable. These results demonstrate that SCI-Mamba maintains effective foreground enhancement and background suppression on previously unseen space imagery, indicating favorable generalization capability under diverse space imaging conditions.

Overall, across synthetic orbital images, darkroom hardware-captured images, real on-orbit imagery, and the independent SPARK 2021 dataset, SCI-Mamba consistently demonstrates effective target enhancement, background noise suppression, structural preservation, and stable illumination correction. In particular, the results on SPARK 2021 further verify that the proposed method generalizes well to unseen space imagery under diverse and challenging imaging conditions.

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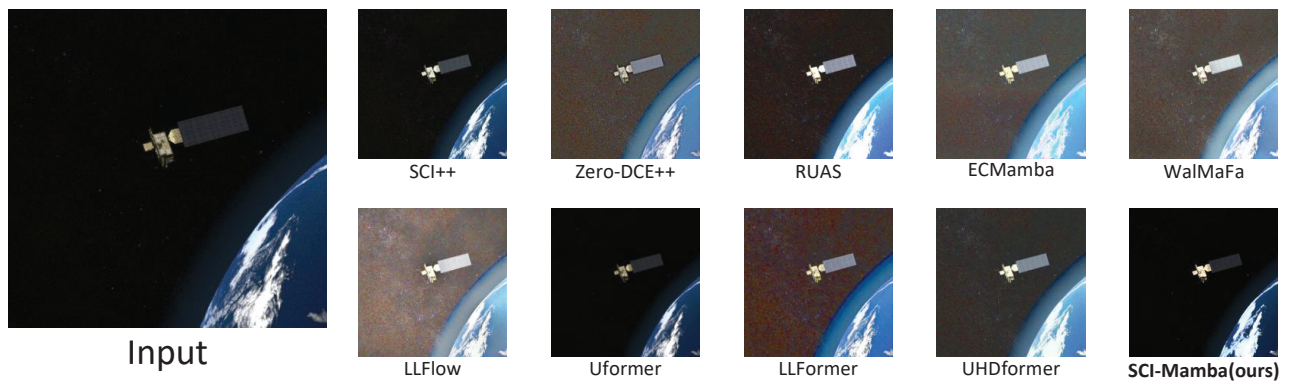


Fig. 4. Visual comparison of different low-light image enhancement methods on the SPARK 2021 dataset.